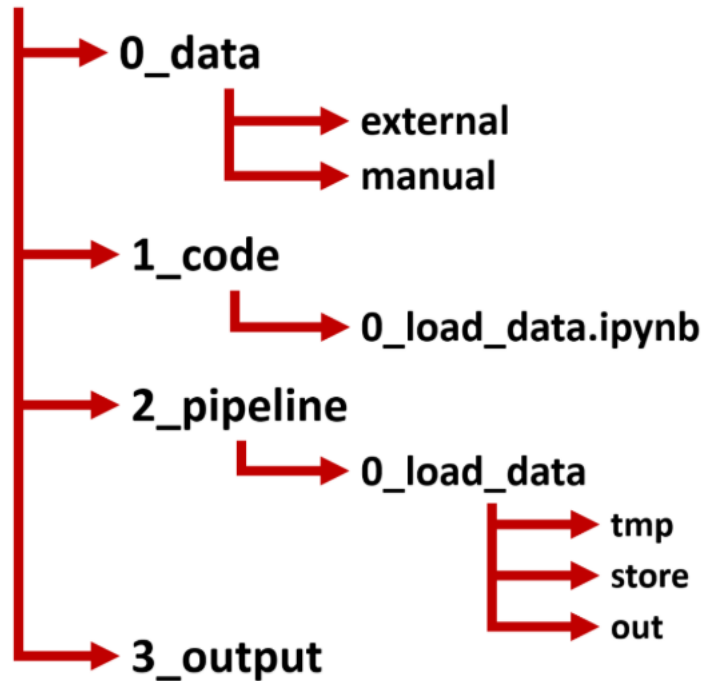


Part1: Creation of Tree Directory

Analyze above structure, and use the same Linux command to create the same directory

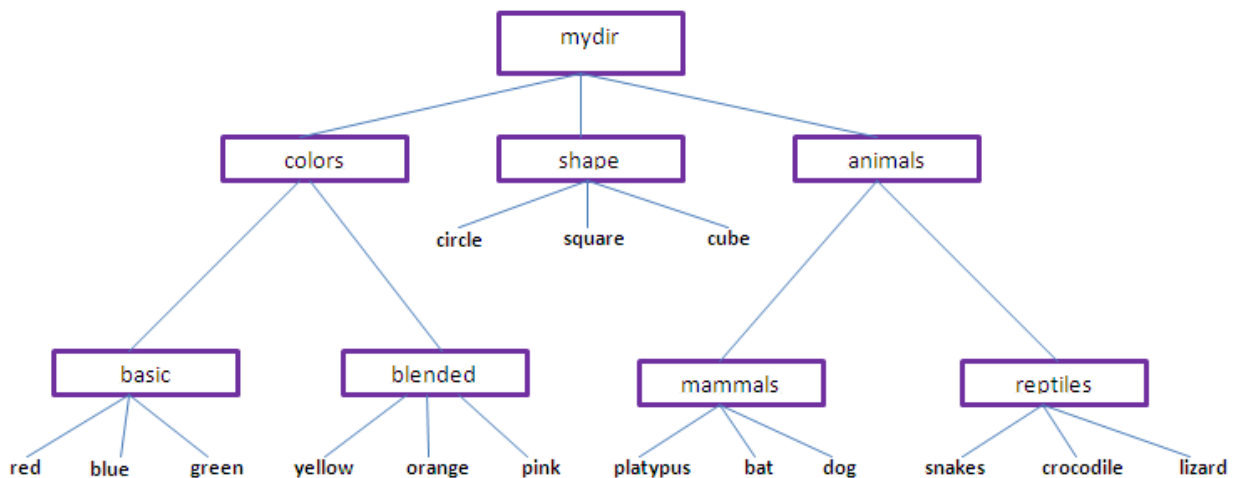
Project Folder



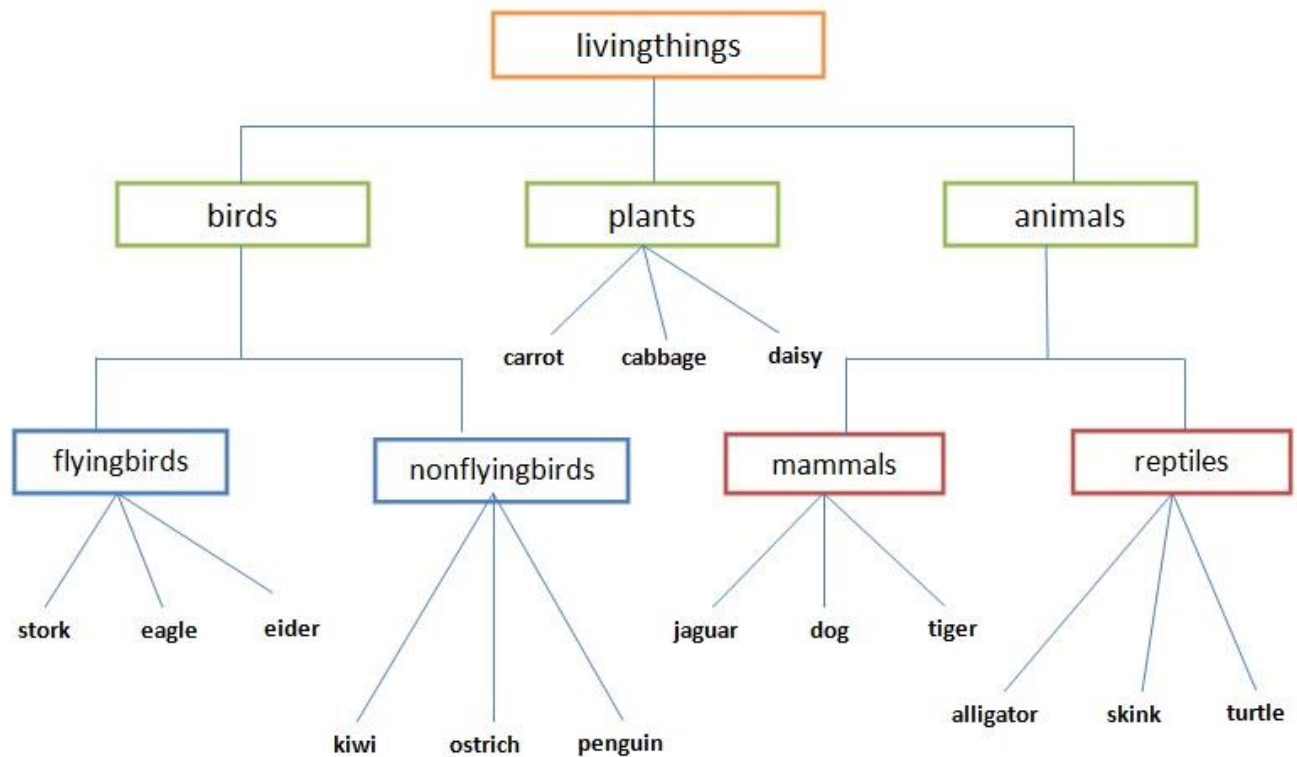
Part 2: File management

I. Write Linux command to create bellow file structure

A. File system I



B. File system 2



Part 3: File permission

1. On jaguar file allow user to have all permission, group permission of read only and other permission of execute only by using symbolic mode.
2. On tiger file allow user to have all permission, group permission of read only and other permission of execute only by using by using absolute mode.
3. On red file allow user to have all permission, group permission of read only and other permission of execute only by using by using absolute mode.

Part 4: Pipes and Filters

1. Write Linux command to Select all file created on specific period “eg dd/mm/yyyy”
2. Write Linux command to Select all file created on specific day and time eg: “2018-10-02 06:37”
3. Write Linux command to Select all file created on “date and time” with corresponding lines number(ID)
4. Write Linux command to Prints all lines that do not match pattern of your choice .
5. Write Linux command to Prints only the count of matching lines of your choice.

Part 5: Network troubleshooting tools

Company XYZ is a mid-sized business with a primary office and several remote branches. Their network infrastructure consists of a mix of servers, including several Linux-based servers for various tasks such as web hosting, file storage, and network services. The primary office network includes a central router, switches, and a firewall. Remote branches are connected via VPN.

Recently, the IT team noticed intermittent network issues, such as slow file transfers, occasional server downtimes, and irregular connectivity drops in remote branches. As part of the network maintenance and troubleshooting team, you are tasked with identifying and resolving these issues.

Questions:

1. Describe the importance of regular updates and how you would update a Linux server.
2. Explain the purpose of the `/etc/hosts` file in a Linux system. How can it be used in network troubleshooting?
3. What is the `ifconfig` command, and how can it be utilized to check network interface configurations? Provide an example.
4. Discuss the role of the `iptables` utility in managing firewall rules on a Linux server. How would you list the current firewall rules?
5. How can the `ping` command be used to diagnose network connectivity issues? Provide an example scenario related to the case study.
6. Describe the use of the `traceroute` command in identifying network path issues. What information does it provide?
7. Discuss the significance of DNS in network operations. How can you test DNS resolution on a Linux system?
8. What is the `netstat` command, and how can it be used to monitor network connections and traffic? Provide an example related to the case study.
9. Describe the functionality of the `tcpdump` utility. How would you use it to capture network packets for troubleshooting?
10. Explain how the `nmap` tool can be used for network scanning and security audits. Provide an example command to scan for open ports on a remote server.
11. How can you use the `ssh` command to securely access remote servers? Describe a scenario where this would be necessary in the case study.
12. List and describe three tools commonly used for network monitoring on Linux. How can each tool be used in network maintenance?
13. You notice high latency on the web server. Explain how you would use `ping` and `traceroute` to diagnose the issue.

14. Describe how to use `netstat` and `ss` to investigate network connections on a Linux server. How can these tools help identify network problems?

15. with relevant Linux command through terminal Generate:

- IP address of <https://companyzyx.gov.rw/>
- Companyzyx network configurations
- The number of hops and packets traveling path required to reach <https://companyzyx.gov.rw/>
- All necessary records (information) assigned to <https://companyzyx.gov.rw/> during deployment and registration for troubleshooting purpose.